

UNIVERSITY PARTNER



Assignment Three – Questions And Answer

6CS012 Artificial Intelligence and Machine Learning

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Submitted on	: <05/17/2025>

1 Long Question:

You are a Data Scientist at eSewa, Nepal's leading digital payment platform.

- Find two major areas where using unsupervised learning could benefit your business the most.
- For each area:
 - Define the problem clearly (e.g., customer behavior clustering, fraud pattern detection).
 - Specify an unsupervised learning process (such as clustering or anomaly detection) and propose algorithms for each process (for example, K-Means, DBSCAN and Autoencoders).
 - Explain in simple words how the findings from the models are useful for eSewa in decision-making and enhancing different features.

Answer: As a data scientist at eSewa- Nepal's largest digital payment service, unsupervised learning has high-impact customer segmentation (personalized marketing) and anomaly detection (fraud prevention).

Customer Segmentation for Targeted Marketing

Problem: To define a clear segment without labeled data, as eSewa has diverse customers who focus on different transactions like mobile recharges, utility payments, and money transfers. Knowing the different behaviors of the customers can help create effective marketing strategies.

Approach: Use of K-Means or Gaussian Mixture Models (GMM) techniques to group users based on transaction habits, number of transactions they make, and use of different services.

Integration: Group of users like: Active users, loyal customers, or high-value clients, can be of use to send targeted messages for marketing, develop loyalty programs or send notifications to increase the engagement of the customers and conversion rates.

Anomaly Detection for Fraud Prevention

Problem: The Traditional fraud systems couldn't detect the fraudulent activities like, account takeovers, unauthorized transfers, when these patterns do not follow the known labels in real time.

Approach: Use of Autoencoders or Isolation Forest techniques to isolate the transactions that do not fit the normal patterns, that helps to catch the new variety of frauds like sudden large transfers from dormant accounts that the traditional rules misses out.

Integration: To enhanced security and protect against the loss of data, any suspicious transaction should be blocked and in cases of high risks, human can review them.

eSewa can find any hidden patterns in customers behavior and detect the fraud by utilizing above mention unsupervised leaning methods to strength the security, improve marketing that supports for better decision-making as well as customer experiences.

2 Short Question

2.1 Overfitting

In the context of machine learning:

- Identify the differences between overfitting and underfitting..
- Describe the obstacles that both of these techniques encounter while constructing models.
- Illustrate your explanation with simple examples (e.g., overfitting a training dataset, underfitting a complex pattern).

Answer: Overfitting occurs when a model learns all the tiny details of the data during training, and perform well on the seen data but in case of unseen data it performs poor.

Underfitting occurs when a model is too simple to learn the underlying features, misses the key patterns of the data during training, and performs poorly on the unseen data during both training and testing.

Why both are problematic:

- The model performs better in the training environment than the testing data which makes it difficult to implement them in the real life.
- Since the model couldn't learn any important relationships in the data, it results in producing unreliable predictions.

Examples: Imagine, trying to predict students' exam scores based on how many hours they study. If model gets too focused on tiny details—like pens color, it might do really well on the data you trained it on, but totally flop when it encounters new students(overfitting). Now, on the flip side, if you try to fit everything with just a straight line, but the actual relationship is more like a curve, your model won't capture that properly. It'll give you poor predictions even on the data it's seen before(underfitting).

2.2 Neural Network Architecture

Describe the main points where traditional neural networks are distinct from autoencoders. Elaborate briefly on different examples where autoencoders have been applied. Detail why autoencoders are useful in these cases and how exactly they assist in solving the respective problems.

Answer: Differences between normal Neural Network architectures and autoencoders

Neural Networks (NNs)	Autoencoders
Traditional Neural Networks are primarily used for supervised learning tasks such as classification and regression.	Autoencoders are designed for unsupervised learning, focusing on reducing data dimensions, learning features, or detecting anomalies.
It typically consist of input, hidden, and output layers that map inputs to predictions.	It have two main parts: an encoder that compresses the input into a lower-dimensional latent space, and a decoder that reconstructs the original input from this compressed form.

It might predict house prices based on features like size, location, and age.	It can be used to compress an image into a simpler form and then reconstruct it, which is useful for tasks like denoising images.
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Application of Autoencoders

1. Fraud Detection: Autoencoders trained on normal transactions learn to reconstruct them accurately, but they struggle with anomalies like fraud, making deviations easier to spot.
2. Image Denoising: They can take a noisy or corrupted image and learn to produce a clean version by filtering out noise.

They help learn efficient representations of data without needing labels. They are great at detecting outliers or reducing noise, which can improve the performance of subsequent tasks.